

# Calendar-Arbitrage Prevention

Convex order, integrated quantiles, and a model-agnostic soft hinge

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## Abstract

Butterfly-freedom keeps a single smile a valid density; calendar-freedom keeps a *surface* consistent across maturities. This note states precisely what that means and how the Vol-Fitter enforces it. Under fixed-forward normalization with deterministic carry, calendar-freedom is four equivalent statements — normalized calls non-decreasing in maturity, the terminal laws increasing in convex order, the integrated upper-quantile curves ordered, and total implied variance non-decreasing at every fixed log-moneyness — and, by Kellerer’s theorem, exactly the condition for the fitted surface to be consistent with *some* martingale. Two enforcements follow. For LQD the integrated upper-quantile curve *is* the asset-share integral every built slice already carries, so the constraint is an elementwise array comparison; expiries are fitted nearest-to-farthest under a soft slack. For the SVI and Multi-Core Sigmoid (MCS) overlays a *model-agnostic* hinge enforces  $w_{\text{far}}(k) \geq w_{\text{near}}(k)$  directly (the local-vol surface needs neither: positivity of its local variance is calendar-cleanliness, Note 04). The note’s case file is the *phantom calendar*: a fixed wide floor grid let a steep short expiry’s linear-wing extrapolation manufacture violations in a no-data region, flattening far fits live on NVDA and SPY; the confinement fix — sample the floor only on the traded range — is reproduced here with the production calibrator (far-fit error 151 vol bp under the wide grid vs 0.0 under the confined one). An explicitly-flagged open problem records what confinement gives up: softer enforcement in the extrapolated region, where options are not yet worthless (remark 2). A worked example takes an inverted-wing pair from a violation of 0.0059 to  $4.6 \times 10^{-7}$ .

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# 1 What calendar-freedom means

**Assumption 1** (Units and carry). Each expiry is normalized by its own forward:  $Y_T = S_T/F_T$  (so  $\mathbb{E}[Y_T] = 1$  for every  $T$ ), quotes live at fixed log-moneyness  $k = \log(K/F_T)$ , and rates, borrow and dividends are deterministic and absorbed into  $F_T$  (Note 06). Comparisons across maturity are made at fixed  $k$  — fixed *forward* moneyness, not fixed cash strike. Discrete cash dividends make the cash-strike statement fail while the normalized one survives; stochastic rates would break the equivalence between calendar spreads and the normalized comparison altogether. The event clock of Note 11 reparametrizes maturity within each slice but preserves the *ordering* of expiries, so the statements below are unaffected.

## Invariant

1. Where data lives, the fitted surface is increasing in convex order: no calendar spread against it.
2. Enforcement is a *soft* hinge — quotes still win a genuine conflict, and the diagnostic reports what slack was used.
3. `enforceCalendar` off, or no previous slice, is byte-identical to independent fits.
4. The constraint is sampled only where *both* slices are pinned by data — the confinement principle (Note 09).

**Theorem 1** (Four faces of calendar-freedom). *Let  $\{Y_T\}$  have unit mean,  $C_T(k) = \mathbb{E}[(Y_T - e^k)^+]$  the normalized calls,  $G_T(\alpha) = \int_\alpha^1 Q_{Y,T}(u) du$  the integrated upper-quantile curves, and  $w_T(k)$  the total implied variances. For  $T_i < T_j$  the following are equivalent:*

- (i)  $C_{T_i}(k) \leq C_{T_j}(k)$  for all  $k$  (no calendar-spread arbitrage);
- (ii)  $Y_{T_i} \leq_{cx} Y_{T_j}$  (increasing convex order);
- (iii)  $G_{T_i}(\alpha) \leq G_{T_j}(\alpha)$  for all  $\alpha \in [0, 1]$ , with equality at  $\alpha = 0$ ;
- (iv)  $w_{T_i}(k) \leq w_{T_j}(k)$  for all  $k$ .

*Proof sketch.* (i) $\Leftrightarrow$ (ii): call payoffs  $(y - e^k)^+$  span the convex functions of  $y$  (every convex function with equal means integrates against the second derivative of the call in strike — the Breeden–Litzenberger pairing), so ordering all calls with equal means is exactly convex order [1, 3]. (ii) $\Leftrightarrow$ (iii): the Hardy–Littlewood characterization —  $\mathbb{E}\phi(Y_i) \leq \mathbb{E}\phi(Y_j)$  for all convex  $\phi$  iff the integrated quantile functions are ordered;  $G$  is the upper-tail version, and  $G(0) = \mathbb{E}Y = 1$  pins the means. (i) $\Leftrightarrow$ (iv): the normalized Black price is strictly increasing in total variance at fixed  $k$ , so ordering prices at every  $k$  is ordering the implied  $w$  at every  $k$  [4].  $\square$

**Remark 1** (Kellerer: why this is the whole story). If in addition each slice is butterfly-free (a valid density — Notes 01–04), then (ii) for all pairs makes the family  $\{\text{law}(Y_T)\}$  a *peacock*, and Kellerer’s theorem supplies a martingale with exactly these marginals [2]. So butterfly-freedom per slice plus calendar-freedom across slices is not merely “no static arbitrage”: it is precisely the statement that the fitted surface is consistent with *some* arbitrage-free dynamic model, even though the fitter never names one.

### Heuristic

Convex order with equal means says the longer law is a mean-preserving spread of the shorter: same forward, more dispersion — exactly what a diffusion does as variance accumulates. Enforcing it between independently-fitted expiries is what stops two good slice fits from implying a negative forward variance between them.

## 2 The LQD-exact enforcement

**Proposition 1** (The asset share is the integrated quantile curve). *For an LQD slice with quantile  $Q(z)$  in the logit coordinate (Note 01), the integrated upper-quantile curve equals the asset-share integral at the corresponding grid point:*

$$G(\alpha) = \int_{\alpha}^1 e^{Q(u)} du = \int_{z_{\alpha}}^{\infty} e^{Q(z)} \Lambda(z)(1 - \Lambda(z)) dz = A(z_{\alpha}).$$

*Proof.* Substitute  $u = \Lambda(z)$ ,  $du = \Lambda(z)(1 - \Lambda(z)) dz$  in the definition of  $G$  and compare with Note 01’s asset-share integral  $A(z)$  — they are the same object.  $\square$

So theorem 1(iii) needs no new computation: every built slice already carries  $A(z)$  on the shared quadrature grid, and the calendar constraint is an elementwise comparison of two arrays — no quantile inversion, no per-strike call comparison — with constraint points automatically dense in the wings (the grid is uniform in  $z$ , not  $k$ ).

The surface is fitted nearest-to-farthest. The constraint subgrid takes every 25th node ( $\sim 320$  points on the 8001-node grid), and each new slice carries the soft squared-hinge slack

$$\sqrt{W_{\text{cal}}} \max(A_i(z_m) - A_j(z_m), 0), \quad W_{\text{cal}} = \text{calendarWeight} = 10^6, \quad (1)$$

keyed on the constraint  $z$ -coordinates, so a slice calibrated on a coarser optimization grid enforces it by Hermite evaluation, bit-for-bit equivalent to the node-indexed form on the native grid.

## 3 The model-agnostic enforcement

The SVI and Multi-Core Sigmoid (MCS) overlays carry no quantile curve, so they use face (iv) directly: a soft hinge on total variance,

$$\sqrt{W_{\text{cal}}} \max(w_{\text{near}}(k_m) - w_{\text{far}}(k_m), 0), \quad (2)$$

on a log-moneyness grid, reading only the previous slice’s `implied_w(k)`. (The local-vol surface needs *neither* enforcement: a positive local variance is calendar-clean by construction — Note 04’s theorem 1(iv) holds along the whole PDE march — so the floor threads only to the parametric overlays.)

The production plumbing is the part that took care. The floor requires the *previous* displayed slice, so `enforceCalendar` threads `prev_display` ascending- $T$  through every path that fits a slice: the single-slice overlay, `fit_and_commit_slice`, the full-surface loop, and the coupled Calibrate workflow all pass the just-fitted display to the next expiry. Wherever the floor is absent (`off`, or the first expiry) the residual block is absent too — byte-identical, test-locked for SVI, MCS, and the display pipeline.

## 4 Case file: the phantom calendar

### Case file: the phantom calendar violation

**Setup.** A steep short-dated slice over a flatter long-dated one — a single-name skew: near SVI  $b = 0.20, \rho = -0.7$ ; far  $b = 0.08, \rho = -0.3$ ; the far expiry's quotes span only  $k \in [-0.25, 0.20]$ . On the data, the far slice sits *above* the near one everywhere: genuinely calendar-consistent.

**Failure mode.** Live SVI overlays on NVDA (Sep-26) and SPY (Jun-27) fitted *flat* with huge RMS whenever the calendar toggle was on.

**Diagnosis.** The floor (2) was sampled on a fixed wide grid  $k \in [-1, 1]$ . With *linear* wings, the steep near slice extrapolates to far higher variance at the grid edges than the flat far slice — so  $w_{\text{near}} > w_{\text{far}}$  out where neither expiry has a single quote. A *phantom* violation, manufactured entirely by two extrapolation laws disagreeing in a no-data region; the hinge then flattened the far fit to cure a problem the data never posed (fig. 1, left).

**Fix.** Confinement: `variance_floor_grid_from` builds the floor grid on the observed quote span  $[\min k, \max k]$  only (41 points; the wide  $[-1, 1]/161$ -point grid survives only as the empty-quotes fallback). The two-curve constraint is sampled only where both curves are pinned by data — the confinement principle of Note 09, of which this incident is the founding instance.

**Verdict (reproduced with the production calibrator, fig. 1).** Under the wide grid the far fit misses its own quotes by 151 vol bp; under the confined grid by 0.0 — indistinguishable from the unconstrained fit. Test-locked, including the assertion that the scenario is consistent on data and inconsistent only in extrapolation.

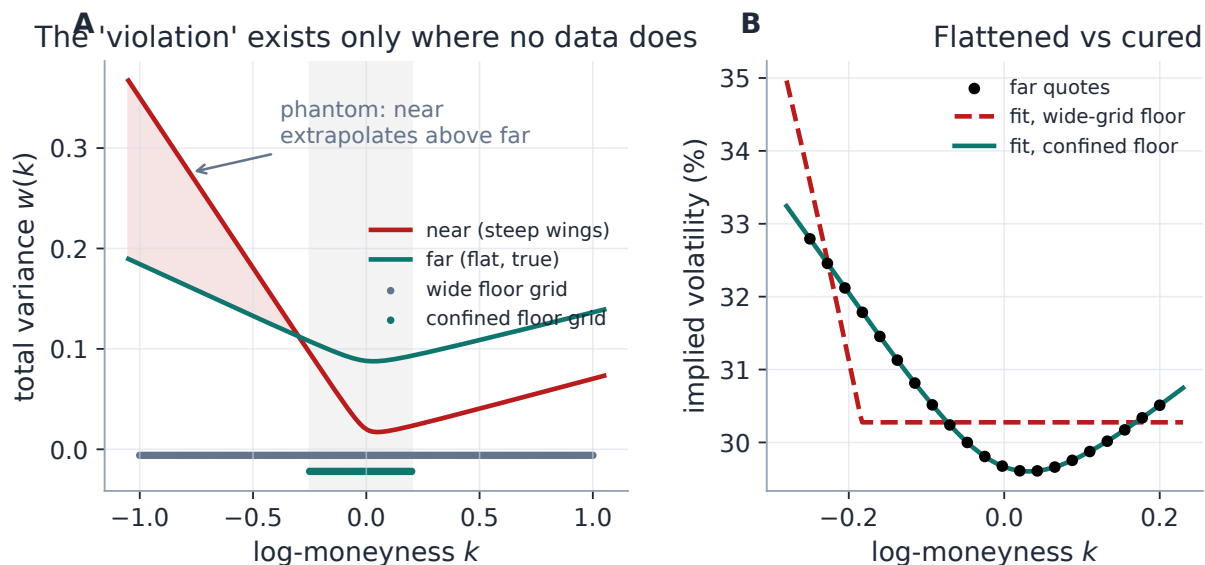


Figure 1: The phantom calendar, reproduced with the production calibrator. Left: on the traded range (shaded) the far slice dominates the near one — no true violation — but the steep near wing extrapolates above the flat far wing exactly where the wide floor grid (dots) samples; the “violation” exists only where no data does. Right: fitted under the wide-grid floor the far smile flattens away from its own quotes (151 vol bp); confined to the traded range it is indistinguishable from the unconstrained fit (0.0 vol bp).

Listing 1: The confinement fix, `variance_floor_grid_from` (verified in appendix B).

```

1 def variance_floor_grid_from(k_quotes, n=41):
2     k = np.asarray(k_quotes, float)      # floor on the TRADED span
        only
3     if k.size == 0:
4         return variance_floor_grid()      # no data at all: wide
        fallback
5     return np.linspace(float(k.min()), float(k.max()), n)

```

**Remark 2** (Open problem: what the extrapolated region still deserves). Confinement is kept — it is the production rule, and the case file above is why. It is not the last word. The confined floor enforces *nothing* outside the traded span, yet the extrapolated region is not uniformly worthless: just past the last quote the options still carry genuine premium, and a calendar inversion there between two *stated* wing contracts (Note 09) is a real inconsistency even if no quote witnesses it. The candidate refinement is a *softer* floor in the near-extrapolation region — the hinge weight decaying with distance past the last quote, or with the options’ remaining time value, so that a gross inversion where options are demonstrably not worthless is still leaned on while the deep-wing disagreement of two extrapolation fictions stays harmless. This is open (flagged 2026-07): any such scheme must be designed jointly with Note 09’s per-model wing contracts and the floor-grid construction of listing 1, and must preserve the case-file property that the NVDA/SPY phantom stays cured.

## 5 Worked example: curing a real violation

Confinement removes *phantom* violations; this example shows the floor doing its real job. Figure 2 fits an *inverted-wing* pair: a high-vol short expiry ( $T = 0.25$ ) over a calmer long one ( $T = 0.5$ ), the term structure an imminent event produces. Fitted independently, the long slice’s total variance dips below the short one’s in the wings — a genuine calendar violation of 0.0059. Refitting under the calendar floor lifts it to dominate everywhere ( $4.6 \times 10^{-7}$ , numerically zero), and the integrated upper-quantile curves come out correctly ordered,  $G_{\text{far}} \geq G_{\text{near}}$  — theorem 1(iii) made visible.

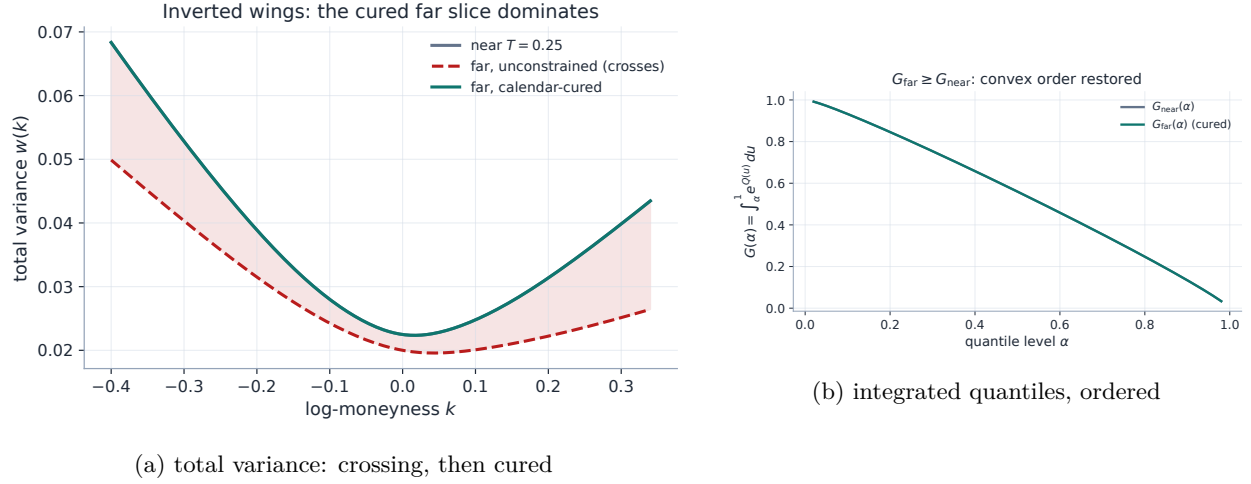


Figure 2: An inverted-wing pair: the unconstrained long slice crosses the short one (violation 0.0059); the calendar floor cures it ( $4.6 \times 10^{-7}$ ).

## 6 What is original here

The convex-order characterization is classical; the contributions are implementation. Recognizing that the LQD asset share *is* the integrated upper-quantile curve (proposition 1) makes the calendar constraint a free elementwise comparison rather than a grid of inverted call inequalities. Extending one calendar treatment to every parametric overlay through the total-variance floor (2) — threaded ascending- $T$  through every calibration path — gives the surface a uniform guarantee regardless of which model draws each slice. And the phantom calendar of section 4 is the incident that crystallized the confinement principle now applied in three other places (Notes 04, 05, 09).

## 7 Traceability

Table 1: Claims in this note and the code/tests that lock them.

| Claim                                              | Object                  | Code anchor                                | Test anchor                                                             |
|----------------------------------------------------|-------------------------|--------------------------------------------|-------------------------------------------------------------------------|
| LQD integrated-quantile constraint; surface repair | con- proposition 1, (1) | calib/calendar.py, models/lqd/calibrate.py | test_surface.py::test_violating_quotes_get_repaired_when_enforced       |
| Model-agnostic floor reads prev slice's $w$        | (2)                     | calib/calendar.py                          | test_overlay_calendar.py::test_variance_floor_reads_prev_total_variance |

| Claim                                                  | Object             | Code anchor                                             | Test anchor                                                                                                                                      |
|--------------------------------------------------------|--------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Floor crushes real violations (SVI, MCS)               | section 5          | models/svi_jw/calibrate.py, models/sigmoid/calibrate.py | test_overlay_calendar.py::test_svi_floor_crushes_calendar_violation, ::test_sigmoid_floor_crushes_calendar_violation                             |
| Byte-identical with no floor / off                     | invariant 3        | models/svi_jw/calibrate.py, models/sigmoid/calibrate.py | test_overlay_calendar.py::test_svi_no_floor_is_byte_identical, ::test_build_display_fit_no_floor_byte_identical                                  |
| Phantom scenario: consistent on data, phantom in wings | section 4          | calib/calendar.py                                       | test_overlay_calendar.py::test_far_above_near_inside_data_but_not_in_wings                                                                       |
| Confined grid fits clean; wide grid flattens           | section 4          | calib/calendar.py::variance_floor_grid_from             | test_overlay_calendar.py::test_wide_grid_breaks_svi_but_data_grid_does_not                                                                       |
| prev_display ascending- $T$ everywhere                 | threaded section 3 | api/service.py, api/workflow.py                         | test_calibration_workflow.py::test_enforce_calendar_threads_prev_into_parametric_items, ::test_enforce_calendar_threads_prev_overlay_for_non_lqd |
| Coupled surface arbitrage-free                         | theorem 1          | api/workflow.py                                         | test_calibration_workflow.py::test_enforce_calendar_surface_is_arbitrage_free                                                                    |

## A Hyperparameter atlas

Table 2: Calendar-arbitrage hyperparameters.

| Knob                            | Default | Role                                                                                                             |
|---------------------------------|---------|------------------------------------------------------------------------------------------------------------------|
| <i>Surfaced</i>                 |         |                                                                                                                  |
| enforceCalendar                 | true    | Master toggle for calendar coupling (all calibration paths).                                                     |
| calendarWeight $W_{\text{cal}}$ | $10^6$  | Soft-hinge penalty strength. <i>Not</i> the graph’s edge-trust calendarWeight of Note 14, which shares the name. |
| <i>Hidden</i>                   |         |                                                                                                                  |

| Knob                | Default   | Role                                                                      |
|---------------------|-----------|---------------------------------------------------------------------------|
| CAL_STRIDE          | 25        | LQD constraint subgrid stride ( $\sim 320$ points on the 8001-node grid). |
| VAR_FLOOR_N_DATA    | 41        | Confined floor grid resolution (listing 1).                               |
| VAR_FLOOR_KMIN/KMAX | $[-1, 1]$ | Fallback wide grid — empty-quotes only.                                   |
| VAR_FLOOR_N         | 161       | Wide fallback grid resolution.                                            |
| floor tolerance     | 0         | Hinge slack offset in the floor builders.                                 |

**Performance.** The LQD constraint is an array slice and subtraction on the already-built asset share — effectively free; its Jacobian rows are  $-\partial A/\partial \theta$  on the active constraints, supplied by the analytic Jacobian of Note 01. The model-agnostic floor is 41 evaluations of the previous slice’s  $w(k)$ . Confinement *reduces* work by not sampling the extrapolation region.

## B Reference implementation

The two enforcement residuals are one line each, and the confined grid is listing 1. The hinge below reproduces the residual blocks inside the SVI/MCS calibrators exactly (the production rows are inline in each calibrator, not a shared function — deliberately, since each model evaluates its own  $w$ ); listing 1 matches `variance_floor_grid_from` output to machine precision on quoted and empty inputs alike.

Listing 2: The calendar hinge as it appears (inline) in each parametric calibrator, and the LQD asset-share form.

```

1  # SVI / Multi-Core Sigmoid (face (iv)): rows are zero unless the far
    dips
2  cal = np.sqrt(calendar_weight) * np.maximum(cal_floor - w_model, 0.0)
3
4  # LQD (face (iii)): the same hinge on the asset share A(z)
5  cal = np.sqrt(cal_weight) * np.maximum(cal_floor - slice_.
    asset_share_at(cal_z), 0.0)

```

## References

- [1] V. Strassen. The existence of probability measures with given marginals. *Ann. Math. Statist.*, 36(2):423–439, 1965.
- [2] H. Kellerer. Markov-Komposition und eine Anwendung auf Martingale. *Math. Ann.*, 198:99–122, 1972.
- [3] P. Carr and D. Madan. A note on sufficient conditions for no arbitrage. *Finance Research Letters*, 2(3):125–130, 2005.
- [4] J. Gatheral and A. Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014.